



CORE PROJECT DOCUMENTATION PACK

Language & Data Strategy

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M A D HALO Technologies



South African languages, particularly isiZulu, Sepedi, Tshivenda, and Xitsonga, are under-resourced in most mainstream AI systems compared to English. Building genuinely useful support for them requires a deliberate strategy — not just switching an interface's display text.

Language Selection

Users choose a preferred language during onboarding, from the currently supported set: English, isiZulu, Sepedi, Tshivenda, and Xitsonga. This selection determines the default language Siyani uses to interact with that user.

Language Switching

Users can change their preferred language at any time, and the change takes effect immediately without losing the context of what they were doing (see the Siyani AI Assistant Strategy document for how this is handled technically).

Translation vs. Actual Language Understanding

There's an important difference between:

- **Translation** — converting existing English content or answers into another language, often literally
- **Actual language understanding** — generating responses natively in the target language, in a way that sounds natural and uses correct, contextually appropriate terminology

Literal translation frequently produces awkward or even confusing results, especially for technical or formal topics (education requirements, government processes) where the "correct" way to say something in isiZulu or Xitsonga may not be a direct word-for-word translation from English. HI NGE SIYANI's goal is the second approach — genuine language understanding — even though it's harder to achieve.

Terminology / Glossaries

Building and maintaining glossaries of key terms (education terminology, government/public-service terminology) in each supported language is a practical, high-value step. This ensures Siyani uses consistent, correct, locally understood terms rather than inventing inconsistent translations on the fly.

Curated Multilingual Content

Rather than relying only on an AI model's general knowledge, HI NGE SIYANI should build a growing library of curated, verified content in each supported language — covering the most common and highest-value education and public-service topics first. This content becomes the foundation Siyani draws on (see the knowledge/content layer in the Technical & Product Architecture document).



Human / Native-Speaker Validation

Automated language generation, even when fluent-sounding, can still contain subtle errors or unnatural phrasing that only a native speaker would catch. Native-speaker review — of both the glossaries and Siyani's actual responses — should be a standard part of expanding or improving language support, not a one-time check.

User Feedback

Users should have a simple way to flag when a response was unclear, incorrect, or "not how we'd actually say this" in their language. This feedback is one of the most valuable, low-cost sources of language-quality improvement available, and should feed directly into content and glossary updates.

Speech Data for Future Voice Capabilities

Voice support will eventually require speech data (recordings, transcriptions) in each target language. This is a significant undertaking and should be approached carefully and gradually — ideally through partnerships (see below) rather than the project alone.

Data Consent and Privacy

Any collection of user data — including speech data, feedback, or conversation history — must be based on clear, informed consent, with users understanding what is collected and why. Sensitive personal information should be minimised, protected (see Supabase RLS in the architecture document), and never used for purposes beyond what users agreed to. This is both an ethical requirement and a trust requirement — HI NGE SIYANI cannot ask people to trust it with education and government-related questions while being careless with their data.

Evaluation / Testing

Language quality should be tested deliberately and regularly — not just assumed to be working because responses "look" fluent. This includes structured review of accuracy, naturalness, and terminology correctness across each supported language, ideally involving native speakers who were not involved in building the content.

Potential University / Research Partnerships

Universities and language researchers working on South African languages (computational linguistics, translation studies, education) are natural potential partners — for content validation, terminology development, and eventually for more advanced language/speech model work. These partnerships can also lend credibility and rigor that a small team can't build alone.

How We Could Eventually Develop Stronger Models for Under-Resourced Languages

The realistic path is incremental:



1. Start with an existing general-purpose AI model, improved through careful prompting and curated content (RAG) — this is the current and near-term approach.
2. Build a growing, validated dataset of high-quality language content and terminology through the steps above (glossaries, curated content, native-speaker review, user feedback).
3. Only once there is a solid, validated dataset and a clear, specific gap that prompting/RAG can't solve, consider fine-tuning an existing model for specific South African languages — likely in partnership with a university or research institution.
4. Training a large language model entirely from scratch is not a realistic or necessary near-term goal, and this document does not recommend pursuing it. The value is in the data and validation work, which pays off regardless of which underlying model technology is used.